

Arithmetic Sequences and Series



Arithmetic sequences

- 1 Consider the sequence 7, 11, 15, 19, ...
 - a State the common difference.
 - b Write a formula for the n th term.
 - c Find a_{20} .
 - 2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.
 - a Find the common difference and the first term.
 - b Write a formula for a_n .
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Arithmetic series

- 3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \dots$
 - 4 Consider the multiples of 6 strictly between 100 and 500.
 - a How many are there?
 - b Find their sum.
 - 5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?
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Arithmetic means and sequence properties

- 6 Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that 5, $b_1, b_2, b_3, 25$ is arithmetic).
 - 7 Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.
 - 8 Prove that the sum of the first n positive odd integers ($1 + 3 + 5 + \dots$) equals n^2 , using the arithmetic series formula.
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Applications

- 9 A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

- 10** A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.